

Prediction of Lung Cancer Risk through Machine Learning based on Lifestyle Questionnaire Data

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Abstract—Lung cancer remains the leading cause of cancer-related deaths worldwide, and early detection is crucial for improving treatment outcomes and survival rates. Hence, the pursuit of alternative methods for early identification of lung cancer should be a significant consideration, especially in low-resource settings. This work presents an alternative for lung cancer risk recognition based on machine learning. We leveraged two publicly available datasets of lifestyle questionnaires from Kaggle, which were collected, processed, and analyzed to develop predictive models. The data was balanced using the SMOTE technique, and seven classifiers were evaluated: Support Vector Classifier, Logistic Regression, Decision Tree, Random Forest, XGBoost, Stochastic Gradient Descent, and Artificial Neural Networks. The best results were obtained with XGBoost, optimized using the GridSearchCV method. We allocated 70% of the data for training and reserved the remaining 30% for testing. The results obtained by the model for the first dataset included accuracy and F1 score of 96.50%, with precision and sensitivity of 96.51%. For the second dataset, an accuracy of 95.83%, F1 score of 95.83%, precision of 96.27%, and sensitivity of 95.83% were achieved. Moreover, using LIME for local interpretability, we were able to identify the primary influence of unhealthy behaviors such as alcohol consumption, smoking, and obesity on the model's predictions, enhancing our understanding of the factors driving the risk of lung cancer in these datasets.

Index Terms—lung cancer, machine learning, lifestyle, explainability, exploratory data analysis, classification

I. INTRODUCTION

Lung cancer is the leading cause of cancer-related deaths worldwide, accounting for 1.8 million deaths in 2020, which is 22% of all cancer fatalities [1]. Thoracic computed tomography and chest radiography are standard diagnostic methods for lung cancer. In contrast, Magnetic Resonance Imaging and Positron Emission Tomography scans are used for staging to determine cancer spread. Accurate diagnosis requires bronchoscopy and biopsy to identify the histological type [2]. These methods are crucial for diagnosing spread and therapy [3] for advanced lung cancer. However, their use in early-stage disease may be limited due to the lack of specific symptoms, potentially leading to unnecessary procedures.

Early identification of lung cancer plays a crucial role in improving survival rates. However, the absence of symptomatic manifestations in the early stages of this disease poses significant challenges in its early detection [4]. Moreover,

repeated exposure to radiation, especially in medical imaging procedures, poses potential health risks such as genetic mutation induction and increased risk of secondary cancer [5]. Therefore, the benefits of these methods must be carefully weighed against potential risks, especially in individuals who may require periodic evaluations. Additionally, the economic and geographical accessibility of these procedures presents a significant barrier, especially in low-resource communities [6], [7]. These challenges underscore the need to consider more strategic approaches in lung cancer diagnosis, especially in resource-limited environments.

Currently, machine learning (ML) techniques play a crucial role in the healthcare domain. For lung cancer specifically, numerous scientific studies have been conducted from the perspective of ML. The results of studies indicate that ML can potentially improve the early detection of lung cancer [8]–[10]. However, many previous works have limitations. A notable observation reveals that many existing studies focus on black-box algorithms, posing significant challenges regarding explainability and understanding. Insufficiently defined model training protocols are expected to be found, raising concerns about the reproducibility and generalizability of results. Lack of transparency in training methodology can affect the validity of conclusions, as details regarding data selection, hyperparameter tuning, and model validation are essential to ensure the robustness and applicability of implementation in clinical settings. Furthermore, many previous works tend to omit the analysis of intrinsic dataset characteristics. Limited evaluation of data quality, imbalanced class distributions, or lack of attention to potential biases may compromise the validity and applicability of developed models. Interpreting results without a comprehensively analyzing these underlying characteristics may lead to misleading conclusions.

This work proposes a lung cancer risk prediction using ML models based on lifestyle questionnaire data. We compare various classifiers to develop a model with maximum sensitivity and discrimination for high-risk individuals, aiming for early, accurate detection and improved treatment prospects. This methodology emphasizes predictive efficacy and result explainability while addressing data quality, class distribution, and potential biases.

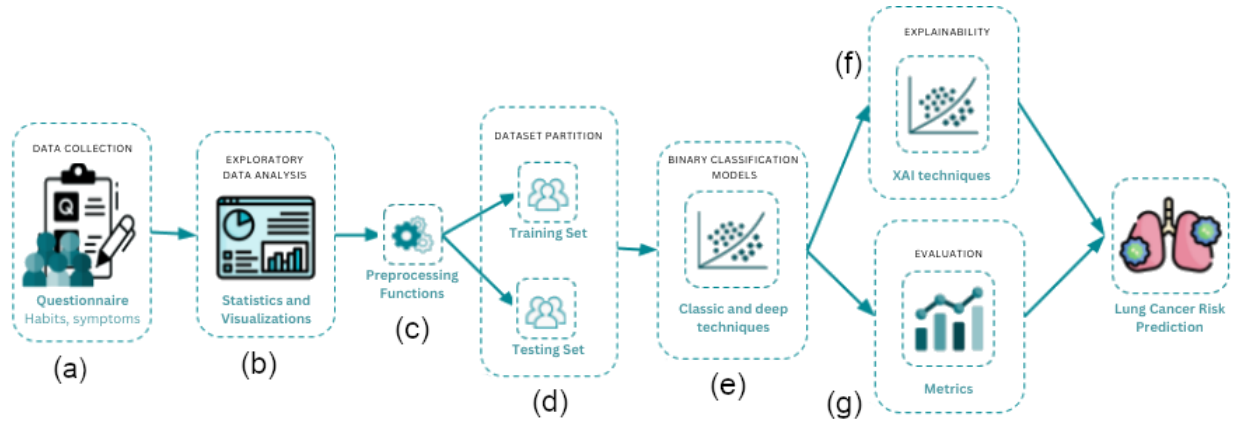


Fig. 1. Overview of the Methodology. (a) The questionnaire data undergo processing and formatting into a tabular structure to facilitate utilization by subsequent algorithms. (b) Subsequently, an exploratory data analysis (EDA) is conducted on the tabulated data. (c) EDA serves as the basis for problem identification and selecting requisite preprocessing functions. (d) The processed data is then partitioned into training and testing sets using the Hold-Out method. (e) Followed by training of binary classification models. (f) These trained models are subsequently employed alongside explainability techniques. (g) The models are evaluated using pertinent metrics to ultimately predict the risk of lung cancer, substantiated by adequate justification.

II. RELATED WORK

Numerous studies have focused on machine learning models employing data collected from specific questionnaires to enhance the early detection of lung cancer. This strategy offers comprehensive insight by integrating information on habits, medical history, and risk factors, key elements for accurate and early detection of this disease.

Some authors have concentrated on developing predictive models for lung cancer prediction using datasets available on Kaggle, exploring various classification ensemble techniques, including XGBoost (XGB) and LightGBM (LGBM). Mamun et al. [11] worked by comparing four classification models: XGB, LGBM, Bagging, and AdaBoost, utilizing 10-fold cross-validation (CV). They utilized an imbalanced dataset available on Kaggle and used the SMOTE oversampling technique. According to their results, the XGB technique outperformed the other three models, achieving an accuracy of 94.42%. Rikta et al. [12] used three classification models: XGB, LGBM, and GBM. They also utilized a dataset published on Kaggle and applied Random OverSampling (ROS) as a balancing technique. The authors further took their analysis by employing Principal Component Analysis (PCA) to select relevant features and used GridSearchCV to obtain optimal model hyperparameters. To explain the results, the authors employed the SHAP tool. Reported results indicate that the GBM classification technique outperformed the other two models with an accuracy of 98.76%.

On the other hand, significant efforts have been made to rely on classical machine learning strategies to predict lung cancer probability. Dritsas et al. [13] employed various classification models, including Naive Bayes (NB), Logistic Regression (LR), Logistic Model Tree (LMT), SVM, SGD, J48, Random Tree (RT), Rotation Forest (RF), Reduced Error Pruning Tree (REPT), Random Forest (RF), AdaBoostM1,

multi-layer perceptron (MLP), and K-NN. They evaluated accuracy, precision, sensitivity, F-Measure, and AUC metrics. The authors applied the SMOTE technique and conducted a 10-fold CV on a dataset available on Kaggle. Reported autor results indicate that RotF demonstrated the best performance among all models, with accuracy 97.1%. Patra R. [8] analyzed various classification models using a dataset on the “UC Irvine Machine Learning Repository”. The author evaluated four classification models: KNN, NB, RBF, and J48. Using Weka, a 10-fold CV was conducted. The results indicate that the RBF algorithm outperformed all other algorithms with an accuracy of 81.3%. Likewise, Ahmad and Mayya [14] used Weka to build decision trees (DT) and RF models to process data from a public dataset on Kaggle. The obtained results were 93.33% of accuracy. Also using RapidMiner, Viera et al. [15] proposed an Artificial Neural Network model from a group of models trained through the CRISP-DM methodology. This model group underwent a series of scenarios where the best model achieved an accuracy of 93%.

Additionally, significant efforts have been made to leverage deep learning strategies as a frontier in research. Thaventhiran & Sekar [16] developed the technique Target Projection Feature Matched with Deep Neural Networks with LSTM (TPFMDANN-LSTM). This model consists of an input layer receiving data and another layer showing results, with two hidden layers in between one for selecting the best features and the other for proper classification using LSTM. The authors tested their model using a dataset from Kaggle and DataWorld, comparing the results with two different methods (LCP-CNN and K-NN). Their proposed model demonstrated better accuracy, fewer false positives, and reduced time consumption.

Overall, these works demonstrate that machine learning has the potential to improve early detection of lung cancer, have a solid methodology, and do not omit dataset characteristics.

However, most works ignore explainability [8], [11], [14], [16]. Omitting explainability in machine learning hinders model transparency and obstructs understanding and trust in model predictions. Unfortunately, only a few works provide an adequate explanation of the datasets used, either during the analysis phase as observed in [13], [15], [17] or by including an explainability report for each prediction generated by the model as seen in [12].

III. METHODOLOGY

The methodology begins by preparing questionnaire data (Figure 1.A), into a tabular format to facilitate their use. Subsequently, exploratory data analysis is conducted (Figure 1.B), from which potential issues are identified and perform necessary preprocessing steps (Figure 1.C). Then the data is split into training and testing sets using a hold-out strategy (Figure 1.D), allowing for robust evaluation of model performance through cross-validation (CV) of the train data. Next, binary classification models are trained (Figure 1.E).

Explainability techniques (Figure 1.F) are applied to clarify the model's decision-making processes, helping to identify biases or areas for improvement. Model evaluation uses metrics like accuracy, sensitivity, and F1 score to ensure objective and valid results (Figure 1.G). Finally, the lung cancer risk prediction is based on the trained models and evaluative metrics. This approach ensures the generation of forecasts supported by solid arguments and transparent justifications, thus substantiating the clinical utility of the conclusions drawn.

A. Data Collection

This study utilized two datasets from Kaggle, both compiled from questionnaires. The first dataset¹ has 309 instances, 15 attributes, and a binary dichotomous label for lung cancer. Regarding the second dataset², it consists of 1000 instances, 23 attributes, and a label related to the level of cancer risk.

In both datasets, some attributes describe habits such as alcohol consumption, obesity, smoking, or passive smoking.

B. Exploratory Data Analysis and Data Preprocessing

We focus on addressing the question: How do habits influence the risk of developing lung cancer? For this purpose, we identified and compared some generalities of both datasets.

1) *Dataset # 1:* All attributes are binary categorical except for age; there are no null or univalent values and 33 duplicate values. Additionally, the dataset is highly imbalanced. The age attribute follows a log-normal distribution (see Figure 3.a), which facilitates how we handle our data. The class attribute "lung cancer" is a binary categorical variable indicating whether there is a risk of lung cancer or not.

The categorical attributes can be divided into habits, mental states, and symptoms. Among the habit attributes, such as alcohol consumption and smoking, it is noteworthy that all female individuals who consume alcohol are at risk of developing lung cancer.

¹<https://www.kaggle.com/datasets/mysarahmadbhat/lung-cancer/data>

²<https://www.kaggle.com/datasets/thedevastator/cancer-patients-and-air-pollution-a-new-link/data>

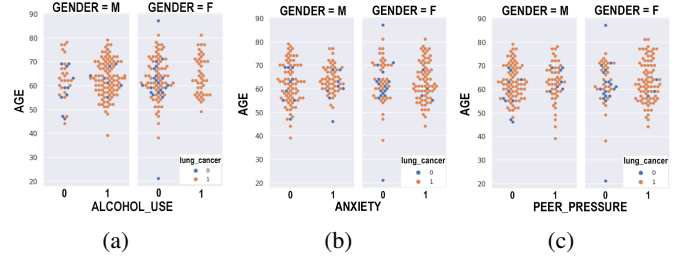


Fig. 2. Swarm plots showing lung cancer risk. (a) Lung cancer risk by age, gender, and alcohol. (b) Lung cancer risk by age, gender, and anxiety. (c) Lung cancer risk by age, gender, and peer pressure. Orange indicates high risk; blue indicates healthy.

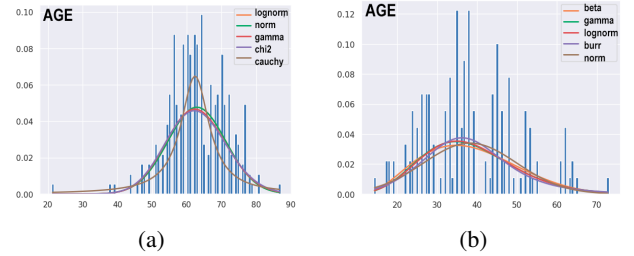


Fig. 3. Histograms illustrating the distribution of the age attribute. (a) Histogram from the first dataset. (b) Histogram from the second dataset.

The majority of male individuals who drink alcohol are also at risk of developing lung cancer, as observed in Figure 2.a. Also, women who present the mental state attributes appear to be at risk of developing lung cancer (Figures 2.a and 2.b). Individuals presenting anxiety and peer pressure appear to be more prone to develop lung cancer (Figures 2.b and 2.c).

2) *Dataset # 2:* All attributes are categorical (scales from 1-9) except for age. No null or univalent values exist, and 848 duplicate values can be highlighted. Despite this, the dataset is slightly imbalanced. Regarding age, it is observed that it fits a beta distribution, as shown in Figure 3.b. The class attribute "level" label is a ternary categorical variable indicating the level of lung cancer risk (low, medium, and high).

The categorical attributes can be divided into habits, environmental, and symptom attributes. Among the habits, such as alcohol consumption, smoking, passive smoking, and obesity, the majority of individuals with a medium or high risk of developing lung cancer are those who consume alcoholic beverages to a greater extent, as depicted in Figure 4.

Additionally, compared to the first dataset, we have two more habits: obesity and passive smoking. Individuals with a higher risk of developing lung cancer generally exhibit a medium or high obesity level. Similarly, the habit of passive smoking poses a high risk of developing lung cancer. Both cases can be observed in Figures 5.a and 5.b.

3) *Comparative Analysis:* In both datasets, we can observe the influence of alcohol consumption, smoking habits, and obesity in individuals at high risk of developing lung cancer.

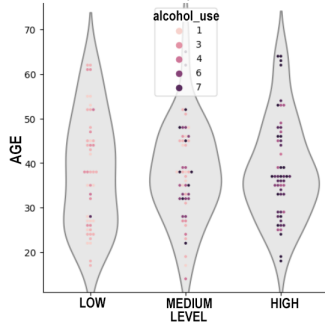


Fig. 4. Violin plot of the risk of developing lung cancer concerning alcohol consumption and age.

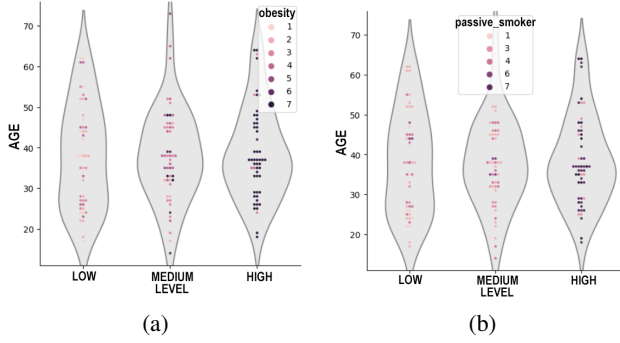


Fig. 5. Violin plots of the risk of developing lung cancer. (a) Risk of developing lung cancer concerning age and obesity. (b) Risk of developing lung cancer concerning age and passive smoking.

We can also find influence from other factors, such as mental states, correlating with this risk.

All these observations will aid in preprocessing the data in both datasets. To address data imbalance, Synthetic Minority Over-sampling Technique (SMOTE) will be applied to generate samples from the minority class.

C. Dataset Partitioning

Once the dataset is balanced, we can select a partitioning method. The method used is Hold-Out with 70%-30% training-testing data. Additionally, we will apply a 5-fold CV for model fitting for the training data. The k -fold CV method is a technique for small-sized datasets, where the main idea is to divide the dataset into k blocks of the same size [18].

We then use one of these blocks for validation and the remaining for training. This process is repeated k times, with the validation block changing to the next one in each iteration until iterating through all blocks. After this process, the result average for each metric to be reported is taken.

D. Classification Models

Once the partitioning phase is completed, we can train a machine-learning model. Below, we explain the different models chosen for the classification task.

Seven classification models, including classical and deep techniques, were applied. The GridSearchCV technique, which

consists of a grid with all the parameters to be tested within a model, was used to obtain the best hyperparameters.

Then, an algorithm is executed to choose a set of hyperparameters to train a model. Through the k -fold CV method, estimated metrics are obtained, which will be used to compare with other sets of hyperparameters and thus obtain the best set.

The selected classification models were Support Vector Classifier (SVC), Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), XGBoost (XGB), Stochastic Gradient Descent (SGD), and Artificial Neural Networks (ANN).

E. Explainability and Evaluation

Once the models are trained using the datasets, we evaluate their generalization power using metrics. These metrics will help us compare each model's generalization performance.

Using the LIME method, we briefly describe the prediction result. Local Interpretable Model-agnostic Explanations (LIME) [19] is a technique that explains predictions of various classifiers in an interpretable manner by learning a local model around the prediction.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

The performance of the models (SVC, LR, DT, RF, XGB, SGD, ANN) within each lung cancer dataset has been computed in Table I for both datasets. For the comparison between each model, we present the results according to accuracy, F1 score, precision, and recall. The metrics were calculated considering the size of each class (weighted).

We can observe that in the case of the first dataset, the Random Forest (RF) model achieved the highest accuracy of 97.20% and F1 score (weighted) of 97.20%. Following this outcome, XGB and LR demonstrated strong performance. For predicting cancer risk within the first dataset, RF performed satisfactorily in accuracy and F1 score.

In the case of the second dataset, the XGBoost (XGB), Logistic Regression (LR), and SVC models achieved an accuracy of 95.83% and an F1 score (weighted) of 95.83%. These models also demonstrated satisfactory performance in predicting cancer risk within the second dataset. Additionally, through LIME, it is possible to explain the prediction results of the different models locally.

For the first dataset, in Figure 6, we have the explainability of a prediction from the RF model, with the 5 most influential attributes being allergy, alcohol consumption, difficulty eating, chronic disease, and coughing. From the feature/value table, we can find that the person's gender is Female (0), they are 74 years old, and they have the condition of peer pressure.

The features that most influenced the XGB and LR model predictions were allergy, difficulty eating, fatigue, alcohol consumption, coughing, chronic disease, and peer pressure.

From these explanations, we can find certain coincidences. Alcohol consumption and coughing are identified as factors influencing the predictions (Figure 6). This aligns with [12], [13], which consider coughing and alcohol consumption among the top 7 characteristics that most influenced the

TABLE I
WEIGHTED METRICS REPORTED BY EACH MODEL
TRAINED ON THE TWO DATASETS.

Results of Dataset I				
Model	Accuracy	F1-score	Precision	Recall
SVC	0.9580	0.9580	0.9580	0.9580
LR	0.9650	0.9650	0.9651	0.9650
DT	0.9091	0.9090	0.9132	0.9091
RF	0.9720	0.9720	0.9720	0.9720
XGB	0.9650	0.9650	0.9651	0.9650
SGD	0.9580	0.9580	0.9584	0.9580
ANN	0.9441	0.9440	0.9444	0.9441
Results of Dataset II				
Model	Accuracy	F1-score	Precision	Recall
SVC	0.9583	0.9583	0.9627	0.9583
LR	0.9583	0.9583	0.9627	0.9583
DT	0.9167	0.9167	0.9167	0.9167
RF	0.9583	0.9581	0.9625	0.9583
XGB	0.9583	0.9583	0.9627	0.9583
SGD	0.9167	0.9158	0.9325	0.9167
ANN	0.9167	0.8546	0.8562	0.8542

local explainability of their predictions. Additionally, we find anxiety states are the most influential characteristics.

In the case of the second dataset: In Figure 7, we observe that the five most influential attributes in the XGB prediction were difficulty breathing, clubbing (abnormal nail bed shape), alcohol consumption, passive smoking, and smoking. Alcohol consumption and clubbing are the most valuable attributes (8 and 9) in predicting a low risk of lung cancer.

From these explanations, we can highlight that alcohol consumption and difficulty breathing are the factors that influence the predictions, which aligns with [12], [13], who consider both attributes among the top 7 influential characteristics in their model decisions, and [14] who consider them as medium-risk factors for lung cancer.

In terms of model performance, our XGB, RF, and LR models, which achieved an accuracy of 96.50% / 95.83%, 97.20% / 95.83%, and 96.50% / 95.83%, respectively, outperformed their counterparts in the related work [11]–[14] except the LR model by Ding et al. (2023) [17]. As we can see in Table II.

On the other hand, ANN achieved an accuracy of 94.41% / 91.67%, which performed similarly or better, in the case of the first dataset, to the model presented by Dritsas et al. (2022) [13] and Viera et al. (2021) [15]. Regarding the second dataset model, its performance is surpassed by the model of Thaventiran et al. (2022) [16].

In summary, we collected the data from the public repository Kaggle, where we dealt with imbalanced and duplicate data. Subsequently, relationships among the data were found through exploratory data analysis.

TABLE II
ACCURACY COMPARISON ON DATASET I (DI) AND II (DII)

Model	Ours	[11]	[12]	[14]	[13]	[17]
DI XGB	0.9650	0.9442	0.9641	-	-	-
DI LR	0.9650	-	-	-	0.9630	0.9677
DI RF	0.9720	-	-	-	0.9520	0.9677
DII RF	0.9583	-	-	0.9333	-	-

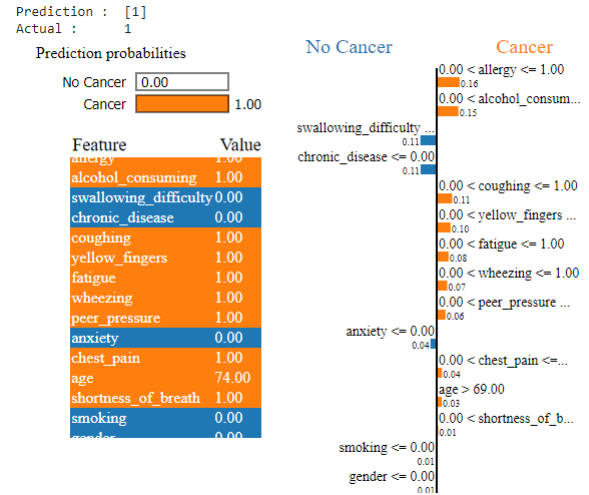


Fig. 6. Feature Importance provided by LIME explainability for the RF model from the first dataset.

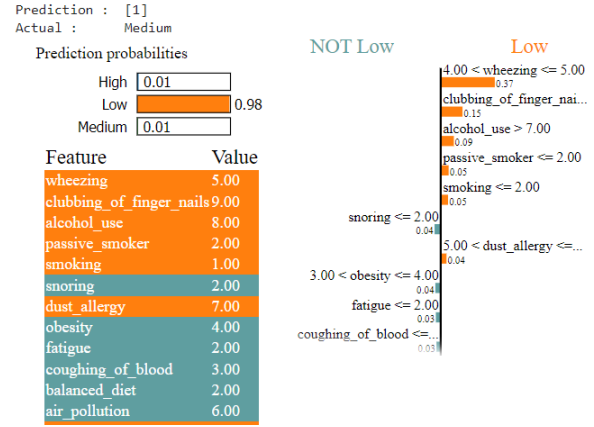


Fig. 7. Feature Importance provided by LIME explainability for the XGBoost model from the second dataset.

We decided to divide the data into 70% - 30% training-testing data. The CV technique and GridSearch were used on the training dataset to find best hyperparameters. The models chosen included SVC, LR, DT, RF, XGB, SGD, and ANN.

Considering the coincidences with the literature [11], [13], [14], [17] and its overall results in both datasets, we decided to propose the XGBoost (XGB) model. The Random Forest (RF) and Logistic Regression (LR) models can also be considered as a second alternative. It is essential to mention that the choice of a machine learning model must always be aligned with the data analysis (EDA); our proposed model should be considered an alternative and not the definitive solution.

The local explainability offered by LIME benefited each model's predictions. It is interesting to note how habits such as alcohol consumption, passive smoking, and obesity significantly influence the predictions.

However, the present study's limitations include the small number of samples in both datasets. Also, our proposed model would need the approval of a specialist doctor to verify its

explainability capacity. We also want to highlight the limited information that can be publicly obtained; accessing data from a hospital or medical institution can sometimes be difficult, mainly due to privacy reasons.

In future work, we hope to obtain more data and features, and also explore more advanced models, such as recurrent and convolutional neural networks. This would give us a broader and more accurate understanding of early lung cancer risk detection. The proposed model can be implemented in public health institutions facing a shortage of pulmonologists to meet the growing demand for patients. Providing support in prevention and prediagnosis, allows healthcare professionals to identify potential risks more early and efficiently. However, challenges include integration into existing healthcare systems, training staff to interpret and use them appropriately, and validating their effectiveness across different populations, ensuring the privacy and security of patient data. Despite these challenges, the use of these tools could significantly improve access to care and diagnosis quality in resource-limited settings.

V. CONCLUSIONS AND FUTURE WORK

Lung cancer is the most common cause of mortality among various types of cancer. Factors such as radiation exposure and economic and/or geographical accessibility to traditional procedures pose a significant barrier to the early identification of lung cancer, especially in low-resource settings.

In this study, we present a methodology for recognizing the risk of lung cancer, which is based on machine learning and the data science process. The best findings indicate that machine learning algorithms, such as XGBoost, Random Forest, and Logistic Regression achieved high performance (Table II), demonstrating their potential for early risk detection of lung cancer. LIME improved the explainability of the results, identifying key factors such as alcohol consumption, smoking, and obesity in the predictions. This highlights the importance of local interpretation techniques to increase confidence in clinical applications. The results obtained by the proposed XGB model for the first dataset included accuracy and F1 score of 96.50%, with precision and sensitivity of 96.51%. For the second dataset, an accuracy of 95.83%, F1 score of 95.83%, precision of 96.27%, and sensitivity of 95.83% were achieved.

Our proposed method is designed to achieve early and accurate risk detection, optimizing medical interventions and improving treatment prospects, while emphasizing the explainability of the results obtained. Despite the promising results of our model, the limitations like the small size of the datasets and the lack of clinical validation restrict our ability to generalize the results. Future work should explore new data collection methodologies, including previously unstudied variables, develop a specific interpretability method for lung cancer, and validate the models with pulmonologists to ensure their applicability in clinical settings. We also aim to build a mobile app to strengthen the connection with rural specialists.

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